Khushaan Virk

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Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science (BASc) in Electrical Engineering — *Expected May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Probability, Signals & Systems, Digital Logic, Embedded Systems

Technical Skills

Programming: Python, C, Embedded C, SQL, SystemVerilog, TypeScript, JavaScript
AI & Cloud: LLM integration, RAG pipelines, agentic AI systems, Supabase, Railway, Vercel, GitHub Actions CI
Embedded & Firmware: ARM (STM32F072), bare-metal C, USB HID, SWD bring-up, ADC, I2C, SPI, UART, PWM
PCB Design: Altium Designer, KiCad, 4-layer stackup, controlled-impedance routing, DRC/DFM, JLCPCB fab and assembly
Tools & Instrumentation: Git, OpenOCD/GDB, ST-Link, oscilloscopes, logic analyzers, SMD soldering, MATLAB

Experience

Software Engineer — Jan. 2026 – Present
Advizr — *Vancouver, BC*

- Raised accuracy on unseen client requests from 32/38 to 37/38 for teammate offers and 13/27 to 27/27 for inbox escalations by leading the swap of five keyword matchers in a production AI agent platform for short model calls that label the text
- Cut CI cost per pull request 18%, $0.67 to $0.55, by benchmarking every heavy job on 2, 4 and 8-core ARM runners, and moved production deploys from every merge to every 3 hours, saving $450 to $600 a month
- Owned uQueue, a clinic appointment search over 3,036 BC Jane subdomains, and moved it to Railway and Supabase
- Ended a go-live outage by tracing it to asyncio's unbounded thread pool draining the 15-connection Supabase session pooler shared with the .NET backend, and capping the scraper's database calls at 5 threads
- Cut BC scrape cycles from 24 to 28 hours to about 4 hours by reclassifying roughly 47,000 routine 404s a run, each tripping a 300-second circuit breaker, as authoritative empty results and adding a per-URL deadline

Sports Instructor / Summer Camp Leader — Sep. 2023 – Sep. 2025
City of Surrey — *Surrey, BC*

- Led structured recreation programs for groups of 15–30, coordinating daily operations across staff and volunteers

Hardware & Firmware Projects

65% Mechanical Keyboard PCB | *Altium, STM32F072, 4-Layer, Bare-metal C, USB HID* — Aug. 2026

- Cut false DRC violations from 194 to 2 by retuning Altium's solder-mask rule to JLCPCB's real 0.1mm capability, which surfaced a diode 0.152mm off grid that would have shipped as a dead key
- Hit 91.0 Ω against a 90 Ω target on the USB differential pair by deriving it by hand from IPC-2141 and routing it at 0.30/0.20mm over solid ground. The board worked on the first spin, 67 of 67 keys
- Proved the 48 MHz clock before pressing a key, with SysTick at 2023 ticks in two seconds, by bringing the board up over SWD with OpenOCD and GDB, then recovered the matrix map from silicon with a tweezer sweep
- Rewrote the firmware bare-metal once QMK had validated the hardware, with no HAL or CubeMX: CMSIS startup, a hand-written Makefile, 1 kHz matrix scan, 5 ms deferred debounce and USB HID via TinyUSB

Pedestrian Dead Reckoning (Phone IMU) | *Python, NumPy, SciPy, pandas* — Jul. 2026

- Counted footfalls from phone accelerometer and gyroscope logs alone, with cadence cross-checked against an FFT of the walk, by band-passing acceleration magnitude to the 0.7–2.1 Hz gait band in NumPy and SciPy and running peak detection
- Measured 5.1 to 6.0 m of inertial drift by loop closure over a 76.2 m route, roughly 7% of distance walked, by estimating gyroscope bias from a stationary window, integrating heading and dead-reckoning an (x, y) trajectory step by step

Field-Following Autonomous Robot | *STM32, C, MOSFETs, I2C, IR* — Apr. 2026

- Built a dual-microcontroller robot with MOSFET H-bridge motor drivers and opto-isolated control lines for electromagnetic noise reduction during inductive sensor readings
- Implemented magnetic field detection using pick-up inductors to autonomously follow a guide wire, with a third sensor for intersection detection and pre-configured path navigation
- Integrated VL53L0X Time-of-Flight ranging over I2C for real-time collision detection and avoidance in autonomous mode

Reflow Oven Controller | *Op-Amp, ADC, SSR, FSM, Python* — Feb. 2026

- Held ±3°C over 25–240°C by designing the thermocouple front end with LM335 cold-junction compensation and OP07 gain
- Built a Python serial data acquisition framework around the firmware: real-time strip-chart plotting and automated Excel report generation over UART at 115200 baud, used as the verification harness during firmware tuning